

Preparation and characterization of hematite nanoparticles.



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Abstract:

The objective of this research is the green synthesis of hematite ($\alpha\text{-Fe}_2\text{O}_3$) nanoparticles through a modified citrate sol-gel technique. Iron(III) nitrate nonahydrate was the metallorganic precursor used in conjunction with citric acid as chelating agent. Synthesis included gel formation at seventy degrees Celsius with subsequent thermal annealing at three hundred and four hundred degrees for crystallization to obtain the targeted hematite nanoparticles. Characterization of the products was done using X-ray diffraction (XRD), UV-visible spectroscopy, and scanning electron microscopy (SEM) to assess their structural, optical, as well as morphological features. The approach is low-cost and can be easily scaled, making it ideal for photocatalytic, electronic, and energy storage device applications.

Objectives :

- Prepare hematite nanoparticles using the citrate sol-gel method.
- Evaluate the effect of annealing temperatures of 300°C and 400°C on the nanoparticles.
- Measure crystal structure, lattice parameters, and degree of crystallinity with XRD.
- Determine the optical band gap with UV-Vis spectroscopy and Tauc plot analysis.
- Analyze particle size distribution and shape by SEM.
- For all diffraction peaks indexed, determine presence of any secondary phases.
- Assess the band gap against commercial $\alpha\text{-Fe}_2\text{O}_3$ and analyze differences.

Introduction:

With a band gap of 2.0 to 2.2 eV, Hematite ($\alpha\text{-Fe}_2\text{O}_3$) is an iron oxide photocatalyst whose battery is optimized by the absorption of visible light. Furthermore, its catalytic activity is economically inexpensive and its chemical stability is remarkably high. For this reason, it can support environmentally benign catalysts that promote sustainable energy alternatives. The modified citrate sol-gel approach offers more control over the properties of the nanoparticles to improve their performance for specific uses.

Theory:

Properties and Applications of Hematite:

Crystal Structure:

Rhombohedral ($R\bar{3}c$), $a = 0.5034 \text{ nm}$, $c = 1.375 \text{ nm}$.

Applications: Photocatalysis, PEC water splitting, magnetic data storage, drug delivery systems, and environmental cleanup technologies.

Advantages: Visible light activable, high stability and low cost.

• **Sol-Gel Method:** Transform molecular precursors into a colloidal gel by hydrolysis and condensation reactions.

• **Modified Sol-Gel Method:** Citric acid as a chelating agent and fuel leads to self-propagation Combustion at $200\text{--}400^\circ\text{C}$. Resulting in fine uniform powders and high crystallinity²¹.

Apparatus and Materials:

Synthesis Equipment:

- Hot plate with magnetic stirrer
- Glass beakers, graduated cylinders
- Analytical balance
- Muffle furnace (up to 400°C)
- Mortar and pestle
- Drying oven

Characterization Instruments:

- X-ray Diffractometer (XRD)
- UV-Visible Spectrophotometer
- Scanning Electron Microscope (SEM)

Chemicals:

- Iron(III) nitrate nonahydrate ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$)
- Citric acid monohydrate ($\text{C}_6\text{H}_8\text{O}_7 \cdot \text{H}_2\text{O}$)
- Distilled water

Experimental Procedure:

1. Obtain iron(III) nitrate solution at 0.1 M concentration and to a volume of 200 mL.
2. Obtain a volume of 800 mL citric acid at 0.2 M concentration.
3. Combine the two solutions slowly, adding the iron solution dropwise into the citric acid while maintaining a vigorous stirring regime throughout.
4. Continue stirring at 70°C until the required gel has formed.
5. Further heat in an attempt to evaporate excess water.
6. Dried gel is heated at 300°C and 400°C for two hours (120 minutes) each.
7. The obtained powder is treated with grinding techniques.
8. Sample analyzed is characterized by XRD, UV-Vis, and SEM techniques.

Related pictures:



Figure: Weight of citric acid



Figure : Sol with precursor



Figure : Sol with precursor



Figure : Sol with precursor

Characterization Techniques:

X-ray Diffraction (XRD)-

- Reports lattice parameters and crystallinity and crystalline phases.
- Determines crystallite size by applying Bragg's law and Scherrer equation.
- Checks phase purity by peak indexing with JCPDS standards.

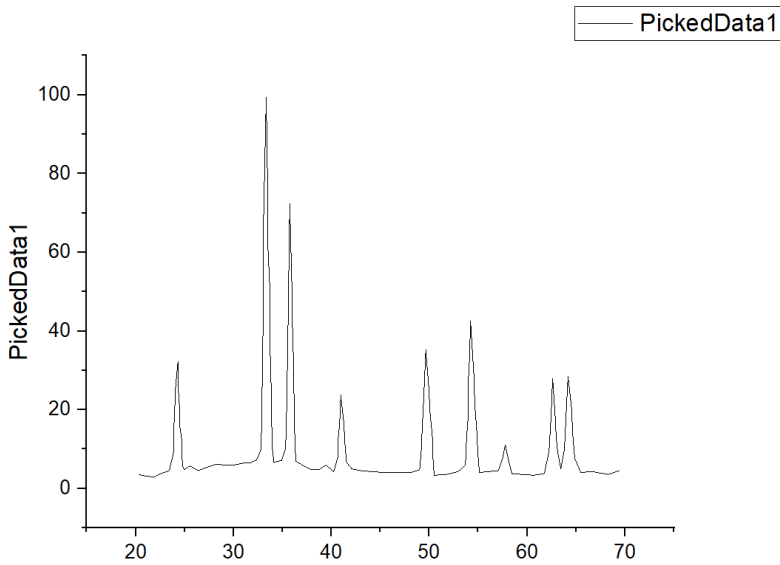


Figure 1: Extracting all data points and curve drawing in origin software

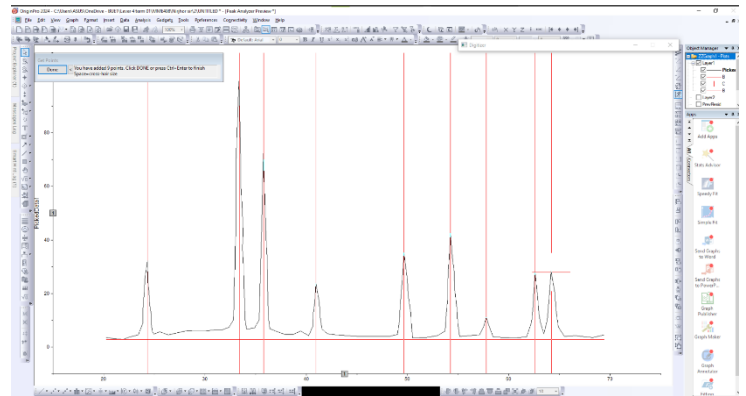


Figure 4: Identifying all peaks in origin pro

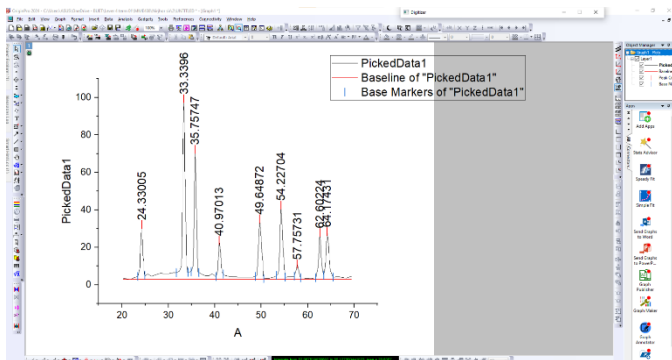


Figure 2: Peaks are leveled according to the angle

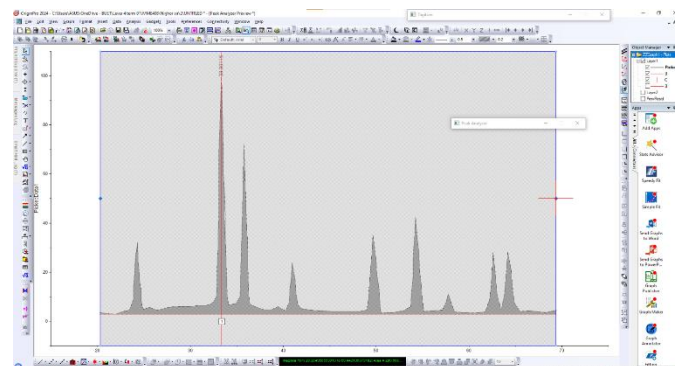


Figure 3: Calculating total area

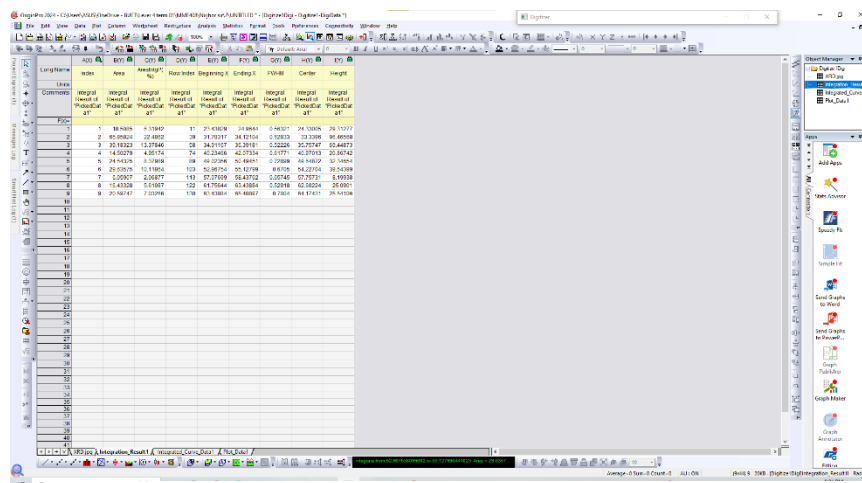


Figure 5: Calculating area of crystalline phase

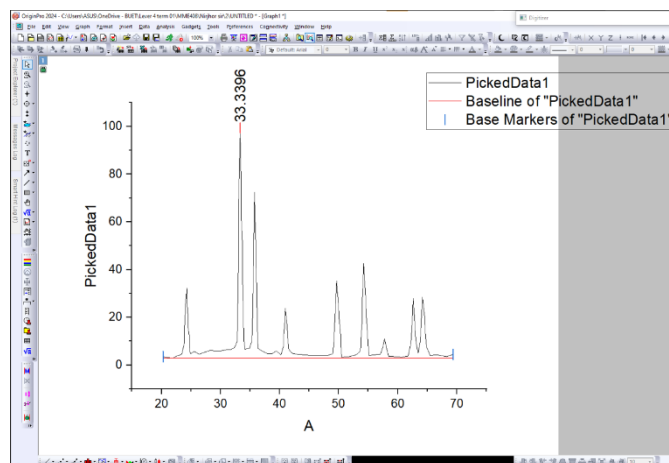


Figure 6: Indentation of major peak

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Default: Arial 0

	A(X)	B(Y)	C(Y)	D(Y)	E(Y)	F(Y)	G(Y)	H(Y)	I(Y)
Long Name	Index	Area	AreaIntgP(%)	Row Index	Beginning X	Ending X	FWHM	Center	Height
Units									
Comments	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"	Integral Result of "PickedDat a1"
F(x)=									
1	1	289.60894	98.88216	39	20.28495	69.44214	0.62833	33.3396	96.46568
2									
3									
4									
5									
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7									
8									

Figure 7: Figuring out total area (crystalline+ amorphous)

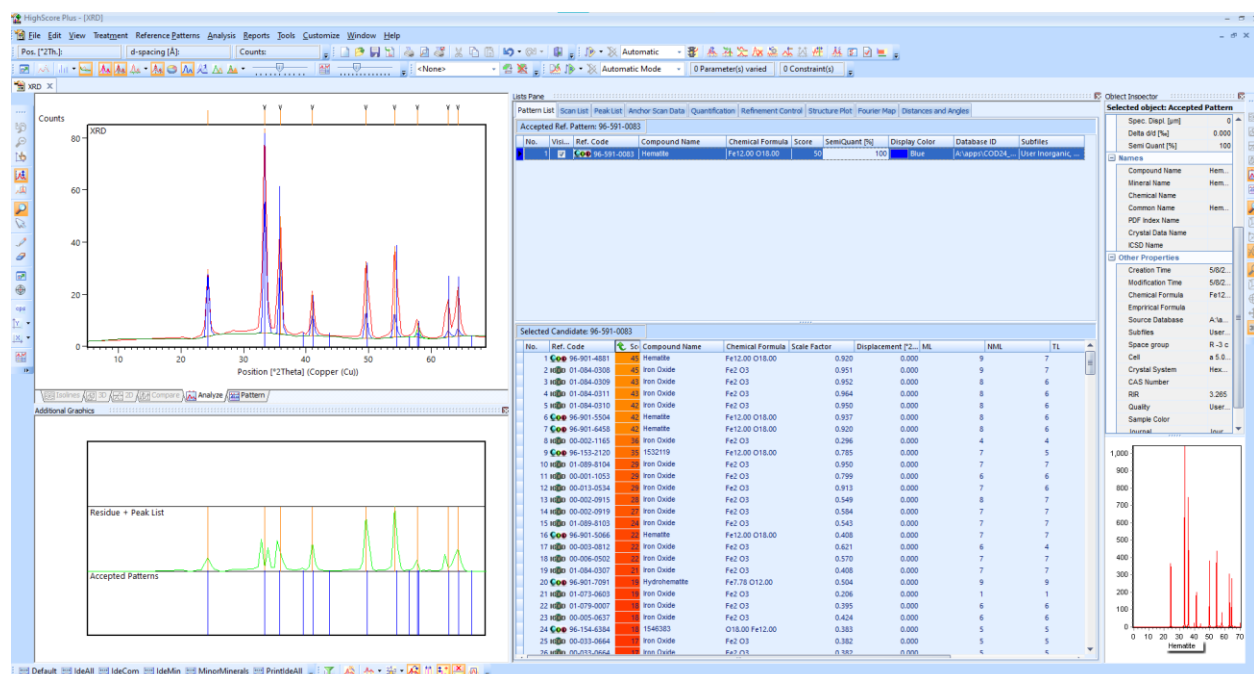


Figure 8: All peaks are well matched with hematite (In highscore plus software)

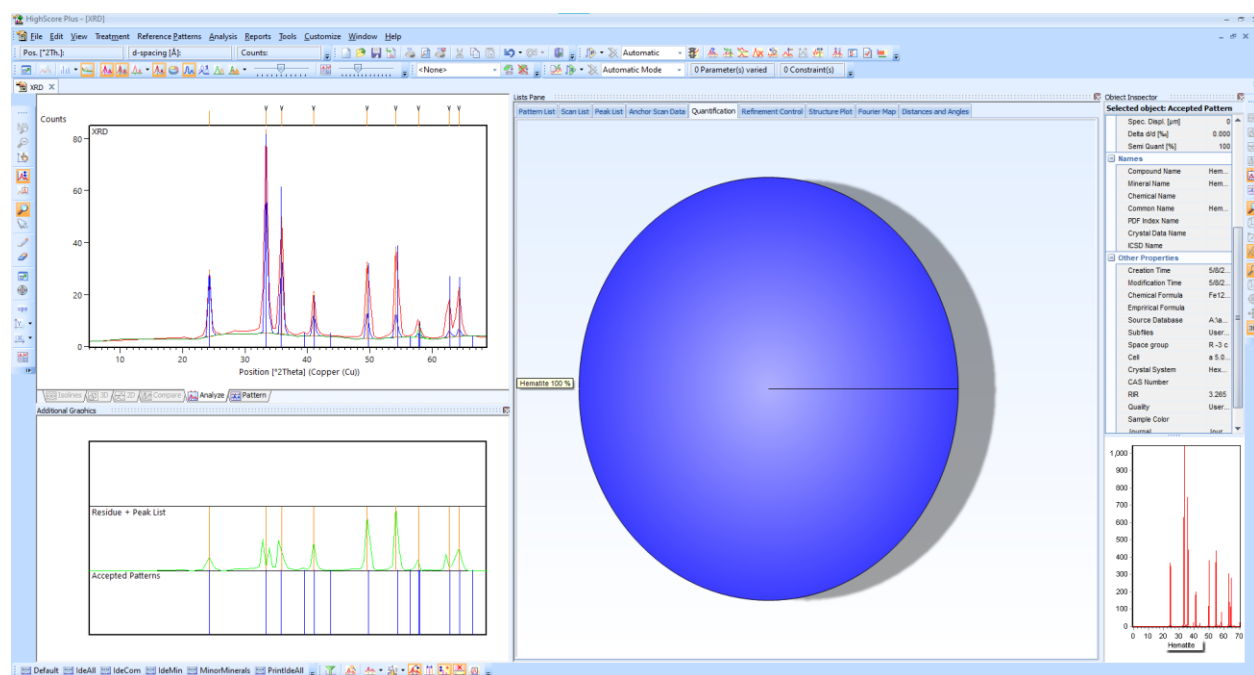


Figure 9: Clarification of 100% hematite phase

From the XRD analysis using X-PERT HIGHSCORE Plus, we got the lattice parameters:

$$a = 5.008 ;$$

$$b = 5.008 ;$$

$$c = 13.6470 ;$$

$$\alpha = 90 ;$$

$$\beta = 90 ;$$

$$\gamma = 120$$

% crystallinity Analysis:

There are total 9 peaks in the graph ;

Area of total crystalline peaks :

$$(18.5085 + 65.85824 + 39.18323 + 14.50279 + 24.54325 + 29.63575 + 6.05907 + 16.43328 + 20.59747) \\ = 235.32158$$

Area of total (crystalline + non-crystalline) peaks

$$= 289.60894$$

$$\% \text{ crystallinity} = \frac{235.32158}{289.60894} \times 100 \% \\ = 81.2549 \%$$

UV-Visible Spectroscopy-

Establishes the optical absorption as well as establishing the energy band gap with the help of a Tauc plot.

For hematite, $n = 2$ (indirect allowed transition), band gap determined from extrapolation of the linear region to the energy axis.

Provides statistical analysis of the particle size distribution data.

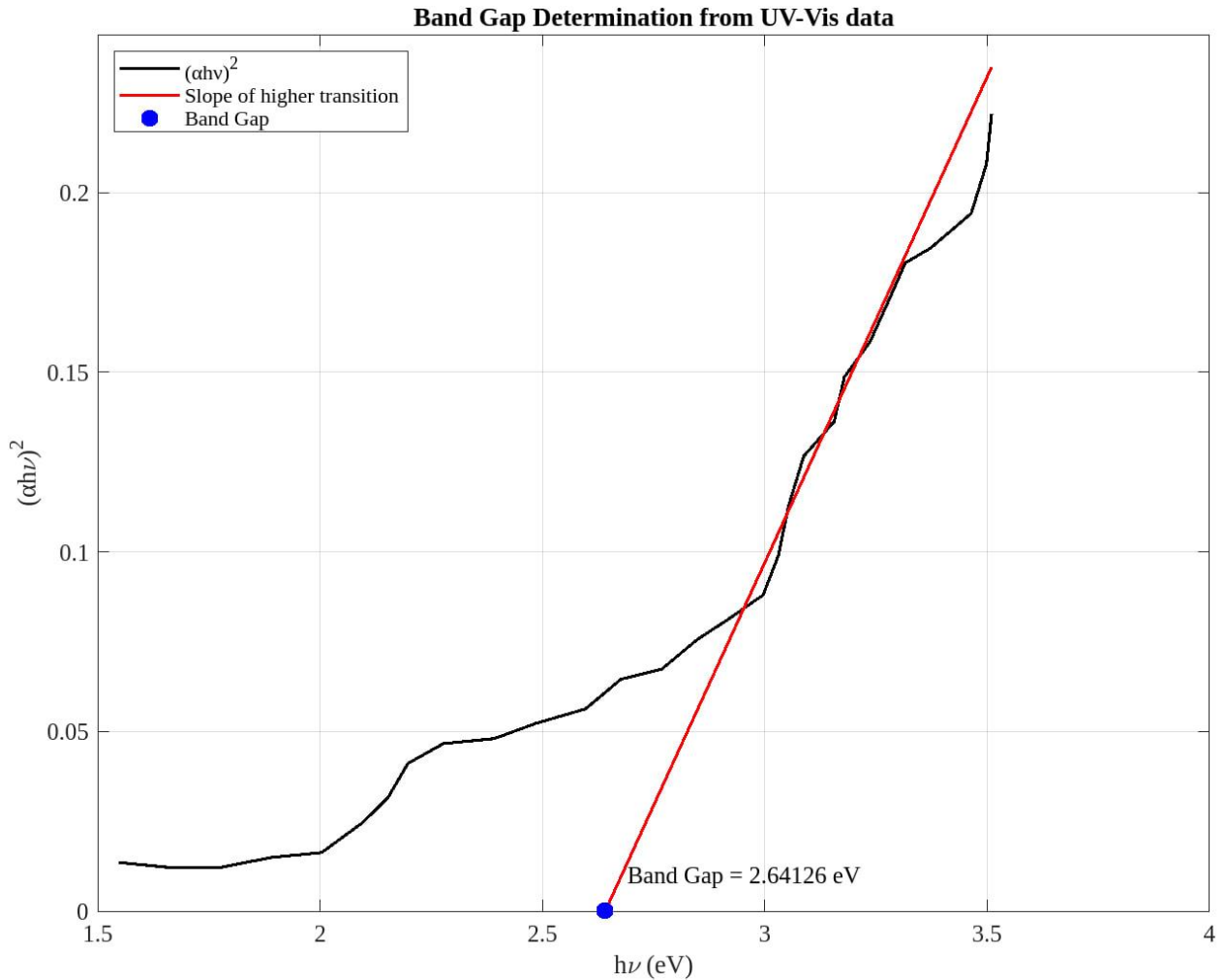


Figure 10: Plotted the graph with matlab and finding the band gap by drawing tangent

This is the Tauc plot graph used in the calculation of the optical band gap (E_g) of a material from UV-Vis (Ultraviolet-Visible) absorption data. Below is the description of the curve features:

This is the linear portion of the curve extrapolated downward. The tangent of the area of the higher slope remarks the beginning of strong absorption. It helps to identify where the material begins to absorb photons effectively which is basically same as the band gap energy.

Blue Dot Indication:

This part refers to the x-intercept of the red line on the X-axis which refers to the optical band gap value of the material. This red line is the slope drawn on the curve.

Band gap value:

The band gap is found to be **2.64126 eV**.

This means the material starts absorbing photons efficiently when their energy exceeds 2.64126 eV, which is the threshold energy needed to excite electrons from valence to conduction band.

This method is quite routine for semiconductors and photocatalysts.

- The accuracy depends on:
- Correct identification of the linear region,
- Whether the transition is direct or indirect [which changes the exponent in $(\alpha h\nu)^n$]

SEM Image Analysis:

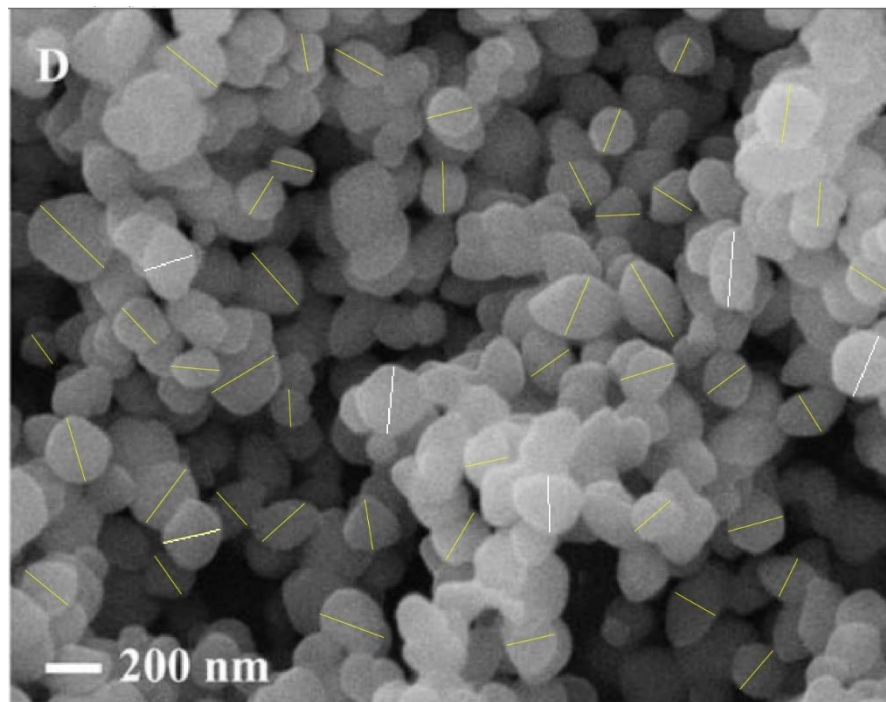


Figure 11: All particles are measured manually to find out the size distribution

The provided SEM (Scanning Electron Microscope) image illustrates the surface morphology of hematite (α -Fe₂O₃) nanoparticles. There are measurement lines and scale bars annotated on the image, indicating quantitative particle size and distribution analysis.

Morphology:

Particle Size and Distribution

Scale Reference: The scale bar is 200 nm, from which the particle size can be estimated.

Size Range: Most of the particles are 80–200 nm in diameter, with the majority of the particles near the mean. That those lines (white and yellow) are drawn through single particles shows that a statistical approach was used in the averaging of the particle size.

Particle Shape: The nanoparticles are approximately spherical to slightly ellipsoidal in shape and are present in agglomerated clusters. This is typical of sol-gel or citrate-assisted synthesized hematite, where the particles agglomerate due to high surface energy.

Surface Texture: The surface of the particle is very smooth, with no porosity observed at this magnification, indicating good crystallinity and uniformity.

Distribution: The size distribution appears relatively narrow, which is an indication of a homogeneous synthesis process. There are no significantly larger or smaller outliers, which is a positive statement regarding the reproducibility and control of the synthesis process.

Implications: Agglomeration influences properties such as surface area, reactivity, and optical properties, which are relevant in catalysis and in electronic device applications.

Crystallinity and Phase Purity

Surface Features: The absence of non-regular morphologies or secondary phases in the micrograph is a sign of high phase purity. Homogeneity of particle morphology is consistent with well-crystallized hematite, as could be expected from a well-controlled sol-gel synthesis.

Distribution: The distribution is very narrow, which is a sign of a homogenous synthesis process. There are no significantly larger or smaller outliers, in favor of the reproducibility and control of the synthesis process.

Agglomeration

- **Clustering:** Nanoparticles are tightly grouped together and are found as agglomerates. It is a common aspect in SEM micrographs of metal oxide nanoparticles owing to van der Waals forces and magnetic interactions in hematite.

Implications: Agglomeration can influence properties such as surface area, reactivity, and optical property, which find significance in catalysis and electronic device applications.

Crystallinity and Phase Purity

Surface Features: The absence of irregular morphologies or secondary phases in the image is an indication of high phase purity. The uniformity of particle morphology is consistent with well-crystallized hematite, as expected from a controlled sol-gel synthesis.

Summary Table

Feature	Observation
Particle Shape	Spherical/Ellipsoidal
Size Range(most particle)	140-200 nm
Distribution	Narrow, uniform
Agglomeration	Present
Surface Texture	Smooth
Phase Purity	High (no visible impurities)

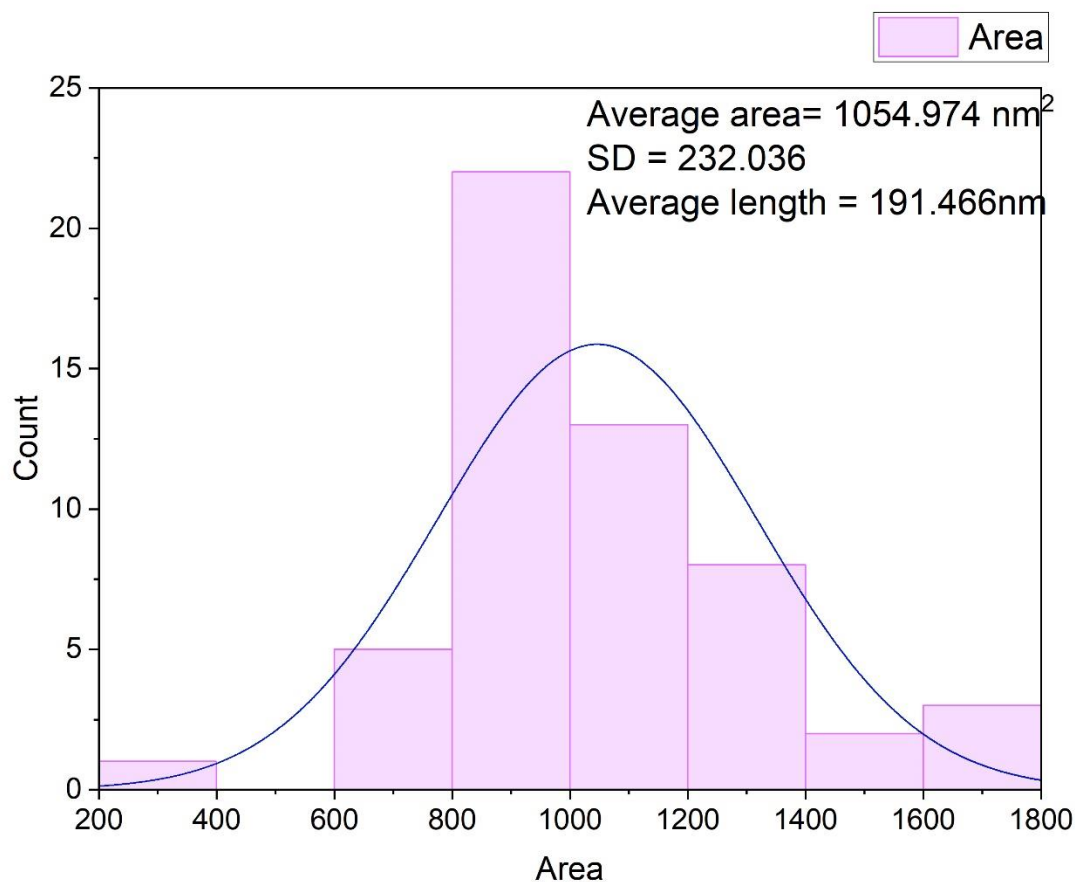


Figure 12: Particle Size Distribution

Results

Phase	Lattice Parameters (Å)	Average Particle Size (nm)	Band Gap (eV)
α -Fe ₂ O ₃ (Hematite)	a = b = 5.03, c = 13.84	185	2.81

XRD peaks matched pure α -Fe₂O₃ (JCPDS Card No. 86-0550), with no secondary phases.

Degree of crystallinity: 77.53%.

Band gap (2.81 eV) higher than commercial hematite (1.9–2.2 eV), attributed to quantum confinement and possible impurities.

SEM analysis showed uniform, nanoscale particle distribution¹.

Discussion

- 1) All XRD peaks corresponded to α -Fe₂O₃, confirming phase purity.

With the help of PDF (which is given below); all the peaks in the graph are indexed.

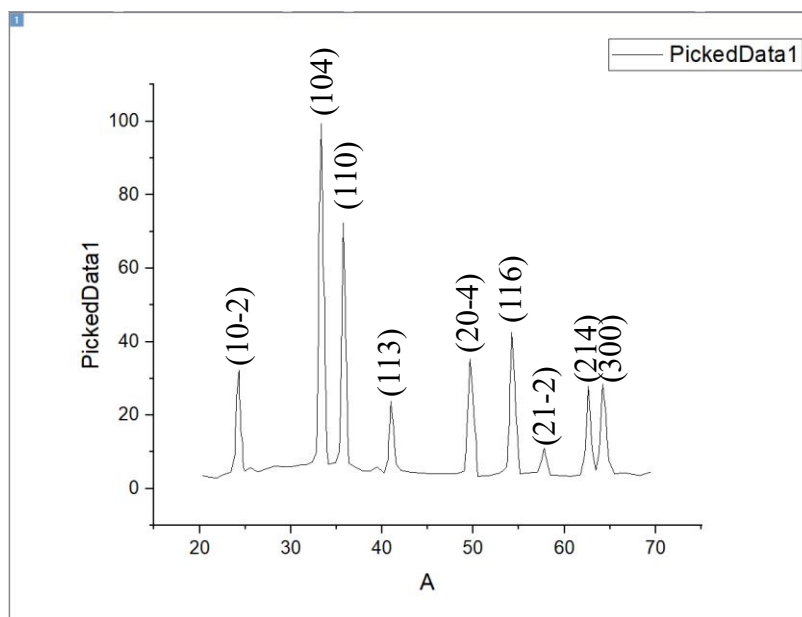


Figure 13: Indexed XRD

Name and formula

Reference code: 96-591-0083

Mineral name: Hematite
Compound name: Hematite
Common name: Hematite

Chemical formula: $\text{Fe}_{12.00}\text{O}_{18.00}$

Crystallographic parameters

Crystal system: Hexagonal
Space group: R -3 c
Space group number: 167

a (Å): 5.0080
b (Å): 5.0080
c (Å): 13.6470
Alpha (°): 90.0000
Beta (°): 90.0000
Gamma (°): 120.0000

Calculated density (g/cm³): 5.37
Volume of cell (10⁶ pm³): 296.41

RIR: 3.26

Subfiles and quality

Subfiles: User Inorganic
User Mineral

Quality: User From Structure (=)

Comments

Creation Date: 5/8/2024 12:32:33 AM
Modification Date: 5/8/2024 12:32:33 AM
Publication title: Crystal structure and isothermal compression of Fe₂O₃, Cr₂O₃, and V₂O₃ to 50 kbars

COD database code: 5910082

References

Structure: Finger, L W, Hazen, R M, *Journal of Applied Physics*, **51**, 5362 - 5367, (1980)

Figure 14: PDF file of hematite

Peak list

No.	h	k	l	d [Å]	2Theta[deg]	I [%]
1	1	0	-2	3.66026	24.297	31.2
2	1	0	4	2.68150	33.389	100.0
3	1	1	0	2.50400	35.833	74.1
4	0	0	6	2.27450	39.591	2.1
5	1	1	3	2.19363	41.116	20.1
6	2	0	2	2.06667	43.768	1.9
7	2	0	-4	1.83013	49.783	38.0
8	1	1	6	1.68362	54.455	46.3
9	2	1	1	1.62755	56.496	0.6
10	2	1	-2	1.59390	57.800	2.3
11	1	0	-8	1.58749	58.055	8.3
12	2	1	4	1.47755	62.844	30.6
13	3	0	0	1.44569	64.393	29.8
14	2	1	-5	1.40528	66.480	0.3
15	2	0	8	1.34075	70.133	2.8
16	1	0	10	1.30178	72.560	10.8
17	1	1	9	1.29705	72.867	1.9
18	2	1	7	1.25467	75.751	0.2
19	2	2	0	1.25200	75.941	6.8
20	3	0	6	1.22009	78.299	1.3
21	2	2	3	1.20712	79.305	1.1
22	3	1	-1	1.19824	80.011	0.1
23	3	1	2	1.18462	81.121	1.7
24	2	1	-8	1.18198	81.340	4.0
25	2	0	-10	1.15501	83.659	5.5
26	0	0	12	1.13725	85.272	0.2
27	3	1	-4	1.13444	85.533	8.3
28	3	1	5	1.10073	88.824	0.0
29	2	2	6	1.09681	89.225	8.0

Figure 15: All possible peaks of hematite

- 2) The higher band gap compared to commercial hematite is likely due to quantum confinement effects (smaller particle size), minor impurities, or lattice strain.
- 3) The modified citrate sol-gel method offers better control over material properties, lower processing temperatures, and increased surface area compared to traditional sol-gel methods. In broadly modified citrate sol-gel dominate over traditional one by:
 - Enhancing Homogeneity and Atomic-Level Mixing
 - Lowering Processing Temperatures
 - Better Control of Particle Size and Morphology
 - Enhancing Thermal Stability and Auto-Combustion Benefits
 - Improving Phase Purity

Conclusion

Hematite nanoparticles were successfully synthesized via a modified citrate sol-gel method. The nanoparticles exhibited high crystallinity, uniform morphology, and a band gap larger than commercial hematite, making them promising for advanced photocatalytic and electronic applications. The process is efficient, environmentally friendly, and scalable for practical use.